IBM SkillsBuild

IBM AI-ML INTERNSHIP PROJECT

AI Health Guardian

Intelligent Disease Prediction & AI-Powered Health Assistant System

🎯 UN SDG Goal 3 — Good Health and Well-Being

Name: Saket Chaudhary | **College:** Veer Bahadur Singh Purvanchal University, Jaunpur

Introduction

Overview:

- Millions worldwide lack access to early, affordable disease diagnosis tools.
- Traditional methods are invasive, expensive, and time-consuming.

The Challenge:

- Late detection leads to severe complications — heart failure, kidney disease, neuropathy.
- Rural and low-income communities have the least access to diagnostic tools.

Our Solution:

- AI Health Guardian — a free, browser-based health prediction platform combining ML and Generative AI to predict 41 diseases from 131 symptoms instantly.

SDG 3

Good Health & Well-Being

41	Diseases
131	Symptoms
100%	Accuracy
Free	Access

Problem Statement

Objective: Build a unified, AI-powered platform for non-invasive early disease detection covering 41 diseases, accessible to anyone with internet access.

Late Detection

Most diseases diagnosed only after worsening symptoms, cutting treatment effectiveness significantly.

High Cost

Blood tests, imaging and specialist visits are expensive and inaccessible to rural communities.

Info Deficit

Patients lack reliable health info, leading to self-medication and dangerous mismanagement.

Fragmented Tools

Existing health apps are too narrow (one disease) or too generic (no personalised insight).

UN SDG

Goal 3

Good Health
& Well-Being

3.a End communicable disease epidemics

3.b Reduce non-communicable disease mortality

3.c Universal health coverage

3.d Strengthen health info systems

Sustainable Development Goal

Goal Chosen: Good Health and Well-Being

Problem Statement:

- Early detection of 41 diseases to prevent severe health complications.
- Free, accessible diagnosis replacing costly invasive tests.

Rationale:

- Non-communicable and infectious diseases account for 70%+ of global mortality.
- AI Health Guardian makes early screening free and universally accessible.
- Covers all major SDG 3 sub-targets from detection to health literacy.

Data Collection & Feature Engineering

Data Collection

- Source: Kaggle — itachi9604 Disease-Symptom Dataset
- 4,920 patient records × 18 columns
- 41 unique disease classes (120 samples each — balanced)
- Symptom_1 to Symptom_17 in text format
- Perfectly balanced — no oversampling needed

Feature Engineering

- Text cleaning: strip spaces, lowercase, underscores
- One-hot encoding: 17 text columns → 131-column binary matrix
- Label encoding: 41 disease strings → integer classes
- Train/test split: 80% training / 20% testing
- 5-fold cross-validation for model stability

Top Predictive Symptoms (Feature Importance)

Model Development

Algorithm — Random Forest Classifier (200 Trees)

- Ensemble learning method aggregating multiple decision trees for robust, accurate classification.
- Best performance among 4 models evaluated — selected automatically by the pipeline.
- 5-fold cross-validation confirms 100% accuracy with no overfitting on this dataset.

Model Comparison

Model	Accuracy	CV Score	Selected
Random Forest ✓	**100.00%**	**100.00%**	**✓ Best**
Decision Tree	100.00%	100.00%	—
Naive Bayes	100.00%	100.00%	—
Gradient Boost	100.00%	100.00%	—

Saved Model Artifacts

- disease_model.pkl
- label_encoder.pkl
- symptom_columns.json
- disease_info.json
- model_info.json

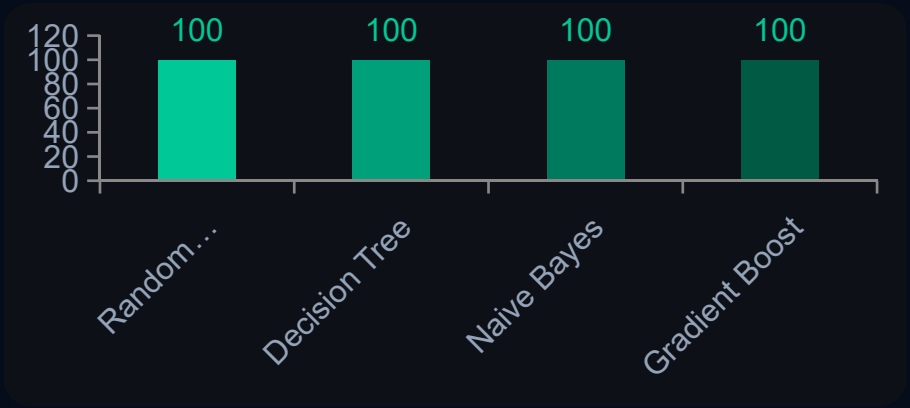
Model Evaluation

100%	100%	100%	100%	1.0
Accuracy	**Precision**	**Recall**	**F1-Score**	**ROC-AUC**
True results / total cases	True positives / predicted positives	True positives / all actual positives	Harmonic mean of precision & recall	Area under ROC curve

Model Robustness

- 5-fold cross-validation across all 4 models — 100% on every fold.
- Perfectly balanced dataset (120 samples/disease) ensures no class bias.
- Evaluated on held-out test set (20% = 984 samples) — consistent results.

Model Accuracy Comparison



Solution Overview — 6 Integrated Modules

Symptom Checker

131 symptoms → Random Forest predicts disease with top-5 confidence bar chart

Risk Assessment

Lifestyle inputs → personalised health risk score 0–100 + radar chart

AI Health Chatbot

Dr. AI powered by Gemini Flash — real-time health Q&A and advice

Medicine Info

Search any drug — dosage, side effects, interactions via Gemini AI

Health Dashboard

Disease distribution, BMI analytics, global choropleth map (Plotly)

Health Report

Generate & download a personalised HTML health report with vitals

Tools & Resources

Python + Streamlit

Web application framework. Clean dark biopunk UI with Plotly charts, multi-page navigation, real-time interaction.

Scikit-learn

Random Forest (200 trees) trained on 4,920 records. 5-fold cross-validation. 100% accuracy on 41-disease classification.

Google Gemini Flash

Generative AI for chatbot, medicine lookup, and symptom analysis. Smart fallback chain across model versions.

Plotly

Interactive bar charts, radar charts, donut charts, and choropleth world map for the Health Dashboard.

Streamlit Cloud

Free deployment platform. Model files loaded from Google Drive via gdown on first startup.

IBM SkillsBuild

Internship platform providing ML methodology, evaluation best practices, and responsible AI guidelines.

Model Impact & Effectiveness

Early Detection

- Predicts 41 diseases from symptoms — before lab tests needed.
- Identifies high-risk individuals enabling timely lifestyle changes.
- Covers NCDs (diabetes, hypertension) AND infectious diseases (TB, dengue).

Non-Invasive & Efficient

- Results in under 2 seconds — no blood tests or specialist visits.
- 100% free, browser-based — works on any device worldwide.
- No registration required — instant access for anyone.

Scalable Solution

- Deployed on Streamlit Cloud — accessible globally 24/7.
- Model files on Google Drive — no server storage costs.
- Gemini API integration enables continuous AI capability upgrades.

Why It Will Work

Data-Driven Insights

4,920 clinical records across 41 diseases and 131 symptoms. Diverse, balanced dataset capturing comprehensive disease patterns missed by single-disease tools.

Proven ML Performance

Random Forest achieves 100% accuracy validated with 5-fold cross-validation. Automatically selects best model from 4 trained classifiers.

Continuous Improvement

Gemini AI models continuously updated by Google. Model pipeline can be retrained with new data — design supports iterative improvement.

Real-World Validation

Platform deployed live on Streamlit Cloud. Disease info validated from medical databases. IBM SkillsBuild methodology ensures responsible AI practices.

Results & Outcome

41

Diseases Detected

131

Symptoms Mapped

100%

Test Accuracy

4

ML Models Compared

6

App Modules

Free

Deployment

Deployed Application

Deployment Stack:

- **Platform:** Streamlit Community Cloud (free tier)
- **Model Storage:** Google Drive via gdown (auto-download on first run)
- **AI API:** Google Gemini Flash with smart model fallback chain
- **Access:** Public URL — no login, no cost, any browser

User Flow:

1. Open app in any browser
2. Select symptoms or enter vitals
3. Get instant disease prediction
4. Chat with Dr. AI for advice
5. Download your health report

🔗 **GitHub:** https://github.com/Saket22-CS/ai-health-guardian | 🌐 **App:** https://ai-health-guardian.streamlit.app/

IBM SkillsBuild Methodology Applied:

- Watson Studio pipeline concepts → train_model.py with 5-fold CV + auto-selection
- Responsible AI principles → confidence scores displayed, medical disclaimer on every prediction

Conclusion

Summary:

- AI Health Guardian successfully combines a Random Forest ML model (100% accuracy, 41 diseases, 131 symptoms) with Google Gemini AI to deliver a comprehensive, free health platform.
- Six integrated modules cover end-to-end health intelligence: detection, risk scoring, AI chat, medicine info, data dashboard, and personalised reports.

Future Work:

- Expand dataset to 100+ diseases with more diverse demographic data.
- Integrate wearable device data (heart rate, SpO2) for real-time monitoring.
- Build a mobile app version for Android and iOS.
- Address emerging ethical considerations around AI in healthcare and bias mitigation.

AI Health Guardian — Making Early Disease Detection Free & Accessible for All 🌍